PHARMACOVIGILANCE ASSESSMENT & CLINICAL CASE STUDY

Individual Case Safety Report (ICSR) Assessment

Role: Drug Safety Associate (DSA) **Task ID:** task5 **Module:** ADR Identification & Evaluation

1. Case Background & Clinical Presentation

A 54-year-old female patient presented to the outpatient clinic with chronic generalized musculoskeletal pain and morning stiffness, diagnosed with active rheumatoid arthritis. Her treating physician prescribed **Methotrexate 15 mg orally once weekly** alongside **Diclofenac 50 mg twice daily** for acute inflammatory symptom management. Folic acid supplementation (5 mg/week) was co-prescribed. The patient had no known prior hepatic or pulmonary diseases.

Six weeks post-initiation of dual therapy, the patient reported to the emergency department complaining of sudden-onset severe upper epigastric pain, melena (dark black stools), extreme fatigue, and dizziness. Clinical laboratory evaluation revealed an acute decrease in Hemoglobin from a baseline of 12.8 g/dL to 8.2 g/dL. Emergent esophagogastroduodenoscopy (EGD) verified acute gastric mucosal erosion and bleeding antral ulcerations.

2. Adverse Drug Reaction (ADR) Identification & Four Elements of Valid ICSR

Under ICH-E2D and global pharmacovigilance standards, a case requires four minimum criteria to qualify as a valid ICSR for regulatory submission:

Criterion	Case Observation & Regulatory Conformance
Identifiable Patient	54-year-old female patient; medical record ID documented; sex, age, and clinical baseline confirmed.
Identifiable Reporter	Attending Emergency Physician / Gastroenterology Consultant (Healthcare Professional - HCP).
Suspect Drug(s)	Diclofenac 50 mg (Primary Suspect); Methotrexate 15 mg weekly (Concomitant / Secondary Suspect).
Adverse Reaction	Acute Upper Gastrointestinal Hemorrhage / Gastric Antral Ulceration with significant Hemoglobin drop.

3. Seriousness & Expectedness Classification

Seriousness Determination: The event qualifies as **Serious** under ICH criteria. It required immediate hospital admission and represents a medically important, potentially life-threatening condition due to active gastrointestinal hemorrhage and severe acute anemia.

Expectedness / Listedness: The adverse event (GI ulceration/hemorrhage) is *expected/listed* in the approved Reference Safety Information (RSI / Summary of Product Characteristics) for Diclofenac (NSAID class). However, concomitant low-dose Methotrexate potentiates mucosal vulnerability, amplifying clinical severity.

4. Causality Assessment (Naranjo Probability Scale)

The Naranjo algorithm was applied to evaluate the causal relationship between Diclofenac administration and the upper GI hemorrhage event:

Naranjo Question	Case Finding	Score
1. Are there previous conclusive reports on this reaction?	Yes (Well-documented)	+1
2. Did the adverse event appear after the suspected drug was administered?	Yes (6 weeks post-onset)	+2
3. Did the adverse reaction improve when the drug was discontinued (dechallenge)?	Yes (Bleeding ceased upon drug hold)	+1
4. Did the adverse reaction reappear when the drug was readministered (rechallenge)?	Not done (Unethical)	0
5. Are there alternative causes that could have on their own caused the reaction?	No primary GI history	+2
6. Did the reaction appear when a placebo was given?	No / Not applicable	0
7. Was the drug detected in blood/body fluids in toxic concentrations?	Not assayed	0
8. Was the reaction more severe when the dose was increased / less severe when decreased?	Not applicable	0
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	No previous exposure	0
10. Was the adverse event confirmed by any objective evidence?	Yes (Endoscopy & Hb Drop)	+1
Total Naranjo Score & Probability Category		**+7 (Probable)**

5. Action Taken & Drug Safety Associate Recommendation

> **Safety Action Plan:** Immediate withdrawal of Diclofenac. Methotrexate temporarily suspended pending complete GI healing. Patient initiated on intravenous proton pump inhibitor (Pantoprazole 40 mg BID) and supportive transfusion. For ongoing pain control, non-NSAID alternatives (e.g., selective acetaminophen or intra-articular steroid injections) are recommended alongside periodic fecal occult blood and CBC surveillance.